

PRINCE SAVSAVIYA

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Professional Summary

Machine Learning Engineer with research experience in retrieval-augmented generation, time-series forecasting, and physics-informed modeling. Skilled in PyTorch, TensorFlow, Scikit-learn, MLOps, and AWS; proven track record improving search precision by 25%, reducing latency by 40%, and publishing peer-reviewed ML research (IF 7.9).

Education

University of California, Riverside <i>M.S. Computer Science – GPA: 3.58/4.0</i>	Riverside, CA	March 2026
Adani Institute of Infrastructure Engineering, GTU <i>B.E. Information and Communication Technology – GPA: 3.6/4.0</i>	Ahmedabad, IN	May 2024

Core Technical Skills

Machine Learning: PyTorch, TensorFlow, Scikit-learn, Deep Learning, BERT, Transformers, LLMs (Hugging Face), Fine-Tuning, Prompt Engineering, RAG, Computer Vision, Time-Series Forecasting, Physics-Informed ML, Model Evaluation, A/B Testing, Hyperparameter Tuning
Languages: Python, C++, Go, SQL, CUDA, Bash, C, Java
MLOps and Cloud: AWS (EKS, EC2, S3), Docker, Kubernetes, MLflow, GitHub Actions (CI/CD), Optuna, Model Monitoring, Prometheus
Distributed Systems and Databases: PySpark, Kafka, Hadoop, FAISS, Elasticsearch, PostgreSQL, ETL Pipelines
Backend and APIs: FastAPI, Django, REST APIs, Microservices

Experience

Graduate Researcher – University of California, Riverside	Riverside, CA	May 2025 – March 2026
- Built ML-ready data pipelines from large-scale multiphysics simulations, enabling surrogate modeling and physics-informed learning at scale; engineered distributed experimentation workflows using Python, Bash, and MPI, improving throughput by 2x. - Designed and evaluated physics-informed ML models including Hamiltonian Neural Networks to reduce solver iterations by 80%; developed automated benchmarking and regression-testing pipelines in cross-functional collaboration with simulation and HPC teams.		
Research Intern – Adani University	Ahmedabad, IN	May 2023 – Dec 2023
- Built a physics-informed ML framework combining PDE ageing models with ensemble regressors (SVR, RF, GBR), cutting MSE by 98% and RMSE by 86% versus a data-only baseline; ingested 200,000+ battery-cycle records with automated feature engineering and Optuna hyperparameter tuning, reducing experiment turnaround by 30%. Publication: <i>Journal of Power Sources</i>, Vol. 627, 2025 (Impact Factor: 7.9) – DOI: 10.1016/j.jpowsour.2024.235771		

Selected Projects

RetrieveX – Semantic Search Engine	Django, BERT, FAISS, AWS	Jan 2025 – Apr 2025
- Architected a hybrid retrieval-augmented generation (RAG) system merging BM25 keyword search with BERT and FAISS vector embeddings, improving top-3 precision by 25% on 150,000 medical documents. - Engineered an asynchronous inference pipeline using Django and REST APIs, optimizing query latency to under 200ms at P95 through end-to-end load testing and model evaluation.		
AI-Driven Infrastructure Autoscaler	LSTM, Kubernetes, Helm, CI/CD	Jan 2025 – Apr 2025
- Designed a time-series forecasting service using LSTMs to predict cluster load, reducing cloud over-provisioning costs by 18%. - Implemented A/B testing for model rollouts and CI/CD deployment to AWS EKS via GitHub Actions; sustained 99.9% SLA under 4x load spikes.		
Risk-Analytics Microservice	FastAPI, Pandas, Docker, AWS	Sep 2024 – Jan 2025
- Designed a real-time analytics microservice with vectorized data pipelines delivering metrics in under 150ms. - Automated deployment via GitHub Actions to AWS ECR and EC2 with Prometheus model monitoring; maintained 99.9% uptime in cross-functional collaboration with stakeholders.		